# Electromagnetic Thoery And Interference

Sem 3 | Watermarked study resource | Generated from local SRM dataset

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===== SOURCE_FILE: sem 3/Electromagnetic Thoery And Interference/Notes/More Study Materials.url =====
CATEGORY: Notes
STATUS: url_reference_only

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CATEGORY: Notes
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===== SOURCE_FILE: sem 3/Electromagnetic Thoery And Interference/Notes/Unit - 5.url =====
CATEGORY: Notes

STATUS: url_reference_only

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===== SOURCE_FILE: sem 3/Electromagnetic Thoery And Interference/PYQs/PYQs Coming Soon.pdf =====

CATEGORY: PYQs

STATUS: ok

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Question Bank

21ECC205T (Electromagnetic Theory and Interference)

1. Given point P ( -2, 6, 3) and vector $\boldsymbol{A} = y\ \boldsymbol{a_x} + (x + z)\boldsymbol{a_y}$, express P and A in cylindrical and spherical coordinates. Evaluate A at P in the Cartesian, cylindrical, and spherical systems.
2. Given tht $\boldsymbol{F} = x^2\boldsymbol{a_x} - xz\boldsymbol{a_y} - y^2\boldsymbol{a_z}$, calculate the circulation of F around the (closed) path shown in the below figure.

3. Determine the gradient of $\boldsymbol{U} = \rho z\ sin\phi + z^2cos^2\boldsymbol{\phi} + \rho^2$.
4. Determine the divergence of $\boldsymbol{V} = \frac{1}{r^2}\ cos\theta\boldsymbol{a_r} + r\ \sin\theta\ cos\ \phi\ \boldsymbol{a_\theta} + cos\theta\ \boldsymbol{a_\phi}$.
5. Evaluate both sides of divergence theorem for field $D = 2xy\ \boldsymbol{a_x} + x^2\ \boldsymbol{a_y}\ \frac{C}{m^2}$, formed by planes $x = 0\ and\ x = 2,\ y = 0\ and\ y = 1,\ z = 0\ and\ z = 2$.
6. Given that D = (5r 2/4)ar (C/m2) in spherical coordinates, evaluate both sides of the divergence theorem for the volume enclosed by r = 1 and r = 2.
7. Determine the curl of $\boldsymbol{W} = \rho\ sin\phi\boldsymbol{a_\rho} + \rho^2z\ \boldsymbol{a_\phi} + z\ cos\phi\ \boldsymbol{a_z}$.
8. For a vector field A, show explicitly that $\boldsymbol{\nabla}.\ (\boldsymbol{\nabla} \times \boldsymbol{A}) = 0$
9. For a scalar field V, show that $\boldsymbol{\nabla} \times (\nabla V) = 0$; that is, the curl of the gradient of any scalar field vanishes.
10. What is coulomb’s law? Write the expressions for Force and electric filed strength due to N point charges.
11. A square plate described by -2 ≤ x ≤ 2, -2 ≤ y ≤ 2, z = 0 carries a charge 12y mC/m 2. Find the total charge on the plate.
12. Point charges 5 nC and -2 nC are located at (2, 0, 4) and (-3, 0, 5), respectively.
(i) Determine the force on a 1 nC point charge located at (1, -3, 7).
(ii) Find the electric field E at (1, -3, 7).
13. Point charges -4 μC and 5 μC are located at (2, -1, 3) and (0, 4, -2), respectively. Calculate the electric filed intensity at (1, 0, 1).

14. Define electric field intensity and write the expression for electric field intensity due to
a point charge, a line charge distribution, a surface charge distribution and a volume charge distribution.
15. Derive an expres sion for the electric field due to a line charge distribution at a perpendicular distance ‘r’ from the line charge. What will be the form of the expression if the line joining the point of interest and the line charge exactly bisects the line charge
distribution. Discuss the special case when the line charge length is infinity.

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16. Find the electric field a distance z above the midpoint between two equal charges ( q), a distance d apart as shown in the figure below.

17. Find E at the point p (1, 5, 2) m in free space if a point charge of 6 μC is at (0,0,1), the
uniform charge density $\rho L$ = 180 $nC/m$ along X-axis.
18. Find the electric field a distance z above one end of a straight-line segment of length L
as shown in the figure b elow that carries a uniform line charge λ. Discuss the special case when $z \gg L$.

19. A circular ring of charge with a radius of 5 m lies in z = 0 plane with center at origin.
If the $\rho L$ = 10 $nC/m$, find the point charge Q placed at the origin which will produce the same E at the point (0, 0, 5) m as shown in the figure below.

20. Give a brief discussion on electric flux and electric flux density. Mention the relation between electric flux density and electric field intensity. Comment on the similar and dissimilar nature of electric field intensity and electric flux density. Write down the expressions for electric flux density due to point charge, line charge, surface charge , and volume charge distribution.
21. Determine the electric flux density D at (4, 0, 3) if there is a point charge -5π at (4, 0,
0) and a line charge 3π mC/m along the y-axis.
22. A point charge of 6 μC is located at the origin and a uniform line charge density of 180nC/m lies along the x-axis. Find (i) D at point A (0,0,4) and (ii) D at point B (1,2,4).

[Page 3]
23. State the Gauss law of electrostatic. Express the mathematical representation of Gauss law in terms of both electric field intensity and electric flux density. From the integral

form of Gaus law derive the differential form of Gauss law.
24. Using Gauss law, find the electric flux density due to a point charge at a distance r from the point charge. Also, find the electric field intensity due to the point charge using both Gauss law and Coulomb's law assuming the point charge to be in free space.
25. Using Gauss law, find the electric flux density and electric field intensity due t o an infinitely long straight wire placed in free space. Assume the line charge density to be $\rho l$. Compare the result with that obtained question 15.
26. Suppose the electric field in some region is found to be $\boldsymbol{E} = kr3\hat{\boldsymbol{r}}$, in spherical coordinates (k is some constant).
a) Find the charge density $\rho$.
b) Find the total charge contained in a sphere of radius R, centered at the origin.
27. If $\boldsymbol{D} = (2r2 + z)\hat{\boldsymbol{x}} + 4xy\hat{\boldsymbol{y}} + x\hat{\boldsymbol{z}}$ C/m2, find
a) The volume charge density at (-1, 0, 3)
b) The flux through the cube defined by $0 \leq x \leq 1$, $0 \leq x \leq 1$, $0 \leq x \leq 1$
c) The total charge enclosed by the cube.
28. Given that $\boldsymbol{D}$ =
$\boldsymbol{5r2}$
$\boldsymbol{4}$ $\hat{\boldsymbol{r}}$ C/m2. Evaluate both side of divergence theorem for the volume enclosed by $r = 4\ m$ and $\theta = \pi\ 4/$ .

Note:
All the vector quantities are represented by bold faced letters.
To make students acquainted with various notations, in some questions the unit vectors are written as ($\boldsymbol{ax}$, $\boldsymbol{ay}$, $\boldsymbol{az}$), ($\boldsymbol{ar}$, $\boldsymbol{a\theta}$, $\boldsymbol{a\phi}$) and in some questions they are represented as ($\boldsymbol{x}$, $\boldsymbol{y}$, $\boldsymbol{z}$),
($\boldsymbol{r}$, $\boldsymbol{\theta}$, $\boldsymbol{\phi}$).

===== SOURCE_FILE: sem 3/Electromagnetic Thoery And Interference/PYQs/Question Bank.pdf =====
CATEGORY: PYQs
STATUS: ok

[Page 1]
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2. Given tht $\boldsymbol{F} = x2\boldsymbol{ax} - xz\boldsymbol{ay} - y2\boldsymbol{az}$, calculate the circulation of F around the (closed) path shown in the below figure.

3. Determine the gradient of $\boldsymbol{U} = \rho z\ sin\phi + z2cos2\boldsymbol{\phi} + \rho 2$.

4. Determine the divergence of $\boldsymbol{V}$ =

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$r2\ cos\theta\boldsymbol{ar} + r$ sinθ $cos\ \phi\ \boldsymbol{a\theta} + cos\theta\ \boldsymbol{a\phi}$.

5. Evaluate both sides of divergence theorem for field $D = 2xy\ \boldsymbol{ax} + x2\ \boldsymbol{ay}$

$C$

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by planes $x = 0\ and\ x = 2,\ y = 0\ and\ y = 1,\ z = 0\ and\ z = 2$.

6. Given that D = (5r 2/4)ar (C/m2) in spherical coordinates, evaluate both sides of the divergence theorem for the volume enclosed by r = 1 and r = 2.

7. Determine the curl of $\boldsymbol{W} = \rho\ sin\phi\boldsymbol{a\rho} + \rho 2z\ \boldsymbol{a\phi} + z\ cos\phi\ \boldsymbol{az}$.

8. For a vector field A, show explicitly that $\boldsymbol{\nabla}.\ (\boldsymbol{\nabla} \times \boldsymbol{A}) = 0$

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